SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA0714 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Chiranjibi Pradhan 192511475(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

TECHNICAL PRESENTATION

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools

Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation


LITE INTERNET
CATION USING THE
DONE BY
GPS Map Camera
Kuthambakkam, Tamil Nadu, India
Eee Department, Kuthambakkam, Tamil Nadu 602105, India
Lat 13.026641° Long 80.015686°
Wednesday, 15/07/2026 12:02 PM GMT +05:30

Satellite Internet communication using the osi layer model

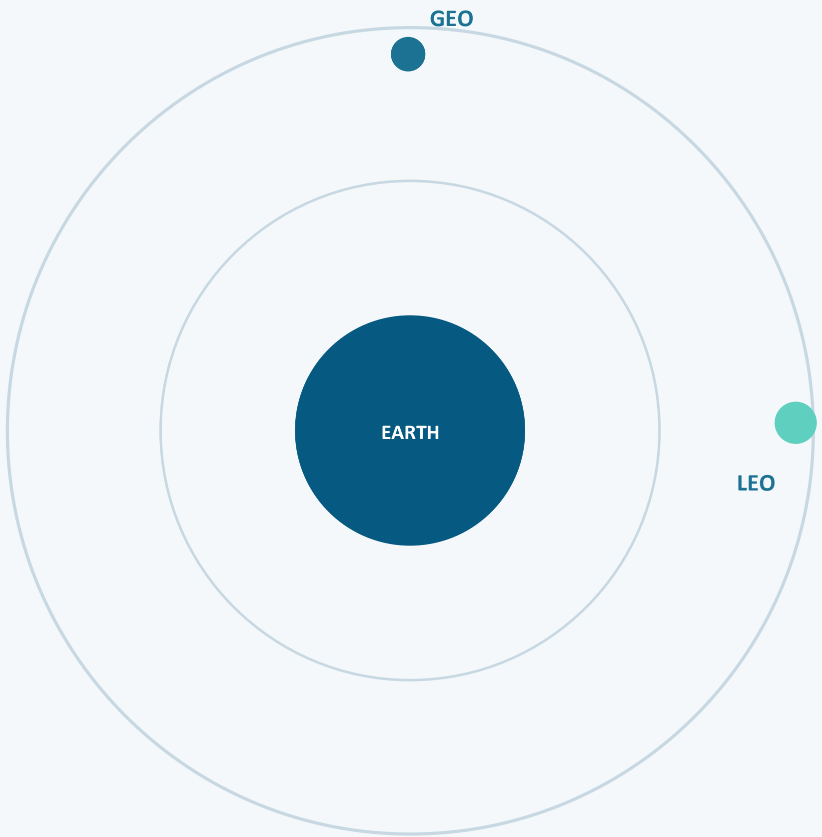
DONE BY
CHIRANJIBI PRADHAN
192511475

What Is Satellite Internet?

Delivering connectivity from orbit, layer by layer

- Delivers connectivity via space-based relays, reaching areas fibre and cell towers can't.
- LEO, MEO & GEO satellites trade altitude for latency and coverage.
- The OSI model still governs the link — every layer must hold up across thousands of km of vacuum.

550 km Typical LEO altitude	**~25 ms** Round-trip latency (LEO)	**35,786 km** GEO orbital altitude



Mapping the OSI Stack to Satellite Links

How each layer stays reliable across a space-based hop

7	**Application**	Web, video & VoIP traffic over the satellite link
6	**Presentation**	Compression & encryption to offset limited bandwidth
5	**Session**	Keeps sessions alive through satellite hand-offs
4	**Transport**	TCP tuned for high latency; fewer retransmits
3	**Network**	IP routing via inter-satellite links & gateways
2	**Data Link**	Error control for signal fade & Doppler shift
1	**Physical**	Ku/Ka-band RF signal, earth-to-orbit

Advantages & Challenges

Reliability trade-offs unique to a space-based OSI path

ADVANTAGES

- Reaches remote areas with no fibre or cell towers
- Rapid deployment — no ground trenching or cabling
- LEO now delivers broadband-comparable speeds
- Resilient when regional disasters sever ground links

CHALLENGES

- GEO delay can exceed 500 ms round-trip
- Rain fade & atmospheric signal attenuation
- LEO hand-offs can drop packets in transit
- Higher cost per subscriber than fibre

Conclusion & Future Scope

Beyond the traditional OSI model

- OSI stays the shared language for reliable transmission — even over a radio link to orbit.
- LEO mega-constellations push tighter Network–Transport integration for real-time routing.
- Next up: AI traffic management, laser inter-satellite links, hybrid terrestrial handovers.

Key Takeaway: Reliable networking is layer-independent — satellite internet proves the OSI model scales from copper wire to low Earth orbit.